SCALE: Dependency-Aware Representation Compilation from Pretrained Transformers to Hierarchical Generative Models

An Adaptive, Mathematically Constrained, and Falsifiable Research Program

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Independent Research / Next-Concept Modeling for TPU
https://github.com/fbalensiefer/Next-Concept-Modeling-for-TPU

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Concept and methods paper — no SCALE benchmark results are claimed

Abstract

Pretrained autoregressive Transformers contain expensive-to-acquire linguistic capabilities, while next-concept modeling (NCM) offers a contrasting coarse-to-fine generative factorization over multiscale discrete codes. We formulate *SCALE* (Size-Consistent Architecture and Latent Exchange) as an *adaptive generative compiler*: given an open-weight teacher and a target NCM resource envelope, it aligns functionally relevant representations into a calibrated intermediate space; designs a rate- and dependency-aware residual-vector-quantized (RVQ) hierarchy; constructs a level-wise conditional generative prior; and applies exact parameter transplantation only where an operator-level equivalence can be justified, otherwise allocating restricted distillation or expanding the target capacity. SCALE is explicitly *not* a universal lossless neural compressor. We give conditional transfer and output-distribution error bounds, an exact characterization of the best factorized within-level prior by conditional total correlation, and capacity and identifiability counterexamples delimiting what no compiler can guarantee. These results establish mathematical coherence under stated assumptions, not universal applicability or performance. We then specify a testable compiler interface, diagnostic gates, controlled baselines, ablation designs, compute accounting, and falsification criteria. The companion public JAX/Flax NCM repository supplies a tested systems starting point, but does not yet implement SCALE, demonstrate learned language quality, or validate transfer or TPU performance. Our contribution is a technically explicit and falsifiable research proposal for *jointly* optimizing representation fidelity, parallel generability, and additional training compute.

Keywords: cross-architecture transfer; neural compilation; next-concept modeling; residual vector quantization; conditional total correlation; model stitching; distillation; JAX/Flax.

1 Introduction

Repeatedly pretraining a generative language model for each target architecture can be prohibitive. Modern Transformer architectures provide the relevant starting point [16]. Yet direct reuse of Transformer weight tensors in an architecturally distinct generator is generally ill-defined: parameter coordinates have permutation and basis symmetries, layers need not correspond one-to-one, and a parallel generator imposes different conditional-independence assumptions from an autoregressive model [1, 15, 18]. Earlier function-preserving network transformations [7], knowledge distillation [9], cross-size feature transfer [5], weight-space merging [8, 14], and Transformer-to-state-space distillation [17] establish meaningful partial precedents. None implies that arbitrary open-weight teachers can be converted into high-quality NCM generators with negligible cost.

NCM, introduced by Kumbong et al. [13], constructs multiscale discrete codes and predicts all codes within a level in parallel, conditioned on preceding levels. The candidate target considered here is an independent TPU-oriented JAX/Flax implementation by Balensiefer [2, 3]; it is a systems adaptation of the original NCM method, not its invention. Crucially, conventional rate–distortion objectives can preserve reconstructive information while placing strongly dependent variables into the same parallel prediction step. This suggests that a useful compiler must co-design the *representation* and the *generative factorization* rather than merely copy weights.

We propose SCALE to connect three elements: (i) cross-size calibrated functional representations; (ii) NCM hierarchies designed for both reconstruction and low residual within-level dependence; and (iii) selective exact transplantation plus bounded, auditable adaptation for incompatible components. The problem is not to prove universal transfer but to specify *when* it is possible, *how* its cost and error will be measured, and *what observations* refute the approach.

Scope and contributions. This paper provides a mathematical problem formulation; propositions with explicit assumptions and proofs; an adaptive compilation algorithm; an information-aware NCM training criterion; a repository integration design; a preregistrable evaluation plan; and explicit negative outcomes. No compiler accuracy, speedup, or transfer result is presented as observed.

2 Related Work and Novelty Boundary

Functional correspondence across models. Model stitching and feature transplantation provide evidence that meaningful representations can align across language-model sizes [5]; CKA offers one descriptive measure of similarity but not a certificate of functional equivalence [12]. MATCHA learns many-to-many layer correspondences between different models, undermining naive fixed-depth matching [15]. Git Re-Basin studies parameter permutation symmetries [1]. We accordingly distinguish *observed activation agreement*, *linear/operator equivalence*, and *end-to-end behavioral preservation*.

Cross-architecture weights and constrained retraining. LS-Merge uses a learned latent weight space for heterogeneous model merging [14]; Transport and Merge uses optimal-transport activation alignment and direct weight fusion [8]; HeteroFusion matches functional module structures under heterogeneity [6]. The Mamba in the Llama reuses portions of pretrained Transformer projections and distills other components [17]. LoRA demonstrates parameter-efficient adaptation, but is not itself a proof of cross-architecture transfer [10]. SCALE borrows these distinctions; it does not claim that alignment, selective reuse, or latent merging is new in isolation.

Discrete hierarchies and parallel dependencies. VQ-VAE established learned discrete representations [20]; residual VQ and structured quantizer dropout have precedent in SoundStream [19]. ResGen addresses generative modeling with RVQ-based tokens [11]. Original NCM introduces the coarse-to-fine text-generator target [13]. Critically, Wen et al. [18] establish conditional total correlation as the information cost of parallel conditional sampling. Our use of this identity is an *application* to learned NCM hierarchy design, not a new discovery of that information identity.

Proposed research distinction. The hypothesis is that *joint* selection of a canonical transfer interface, an NCM hierarchy with low conditional total correlation, and certified selective operator reuse can outperform applying these existing tools independently at matched total compute. This

is a *testable systems-and-methods contribution*, not a substantiated claim of global priority or patent novelty.

3 Problem Setup and Compiler Contract

Let s index a teacher model T_s with parameters W_s, hidden widths d_s, and conditional text distribution $p_s(x \mid u)$ for prompt u and continuation x. T_s may be autoregressive. The target is an NCM generator $S_{\omega,\phi}$ with a codec $q_\phi : x \mapsto c$ (possibly conditioned on u), discrete levels $C = (C_1, \ldots, C_K)$, a prior $q_\omega(C \mid u)$, and a decoder $d_\phi(x \mid C, u)$. Its induced output law is

$$q_{\omega,\phi}(x \mid u) = \sum_C q_\omega(C \mid u)\, d_\phi(x \mid C, u). \tag{1}$$

For a given sequence geometry, the NCM prior factorizes as

$$q_\omega(C \mid u) = \prod_{k=1}^{K} \prod_{j=1}^{n_k} q_{\omega,k,j}(C_{k,j} \mid C_{<k}, u, G), \tag{2}$$

where G denotes lengths, masks, and permitted conditioning metadata. Within-level token predictions may interact through deterministic attention, but sampled code values $C_{k,j}$ do not condition on other newly sampled codes in that level.

Compiler interface. A proposed compilation procedure is

$$\mathsf{Compile}(T_s,\, \mathcal{D}_{\mathrm{cal}},\, \mathcal{B},\, \mathcal{R}) \longrightarrow (S_{\omega,\phi},\, \Pi_s,\, \mathcal{A}_s), \tag{3}$$

where $\mathcal{B}$ specifies output-architecture, storage and latency limits; $\mathcal{R}$ specifies admissible adaptation compute; Π_s stores calibration maps and code geometry; and $\mathcal{A}_s$ is an *audit certificate* recording which operators were proven exactly transferable, approximated, retrained or rejected. "Compiler" means compilation from an accessible model and samples to an executable generative target; it does not mean a training-free algebraic conversion.

A useful scientific objective is the constrained multi-objective program

$$\begin{aligned} \min_{\phi,\omega,\Pi_s,G} \quad & \underbrace{\mathbb{E}_{u\sim\mathcal{U}} D_{\mathrm{KL}}\big(p_s(\cdot \mid u) \,\|\, q_{\omega,\phi}(\cdot \mid u)\big)}_{\mathcal{L}_{\mathrm{gen}}} \\ & + \lambda_h \mathcal{L}_{\mathrm{rep}} + \lambda_{\mathrm{tc}} \mathcal{L}_{\mathrm{tc}} + \lambda_r \mathcal{L}_{\mathrm{rate}} \\ \text{s.t.} \quad & \mathrm{FLOPs}_{\mathrm{adapt}} \le \mathcal{R}, \quad \mathrm{Memory}(S) \le \mathcal{B}_{\mathrm{mem}}, \end{aligned} \tag{4}$$

with additional quality–cost Pareto analysis rather than a single arbitrary scalarized winner. As full-sequence D_{KL} is generally intractable, Section 6 defines practical estimators and reports their limitations.

4 Architecture: An Adaptive Generative Compilation Pipeline

SCALE has six logically separable stages.

4.1 Teacher inspection and calibration

An inspector records model/tokenizer revisions, vocabulary and generation conventions, embedding and attention/operator layouts, normalization, positional encoding, intermediate tensors and shape semantics. A frozen teacher runs on an agreed calibration corpus with prompt/continuation identifiers and recorded random seeds. Multi-depth activations $H_{s,\ell}(u, x)$ and teacher logits are captured; layer pairing is estimated rather than assumed [15]. Unavailable activations or weights limit which paths can be certified.

4.2 Canonical functional intermediate representation

A reusable compiler core defines a latent dimension r, optional module-level types, and a canonical feature interface Z. Per-teacher maps $P_{s,\ell} : \mathbb{R}^{d_s} \to \mathbb{R}^r$ and partial reconstructions $R_{s,\ell} : \mathbb{R}^r \to \mathbb{R}^{d_s}$ are calibrated, e.g. by regularized ridge regression against aligned anchor features. A prototype objective is

$$\min_{P_{s,\ell}} \left\| H_{s,\ell} P_{s,\ell}^\top - Z_\ell \right\|_F^2 + \rho \left\| P_{s,\ell} \right\|_F^2 . \tag{5}$$

The latent interface need not preserve everything the teacher knows: it should preserve task-relevant predictive distinctions, with reconstruction/probe tests, held-out logits and failure flags. Source-specific calibration is permitted and its cost counted. The pretrained compiler *core*, unlike the calibration maps, is held fixed in cross-size experiments.

4.3 Selective operator transfer

An operator analyzer represents candidate transforms as typed computational subgraphs, not arbitrary arrays. For a linear operator W_s, a canonical approximation $\widetilde{W}_s = P_s W_s R_s$ is admissible only with compatible input/output maps, rank checks and a measured residual on independent anchors. Exact transfer requires an algebraic identity on a specified domain, such as Proposition 1. Nonlinear attention blocks, layer normalization, positional schemes and different vocabulary heads are *not* declared exact by applying this formula blindly. They are assigned a bounded alignment/distillation budget, possibly low-rank adaptation [10], or an explicit rejection status.

4.4 Dependency-aware discrete hierarchy design

A codec maps canonical target representations z into a sequence of coarse-to-fine RVQ codes. For levels $k = 1, \ldots, K$, let V_k be codebook sizes and n_k valid positions; masked resampling aligns levels to a common representation grid. SCALE chooses code schedules, allocations and codebook initializations subject to a rate budget while favoring a hierarchy where early levels explain shared information that would otherwise remain correlated within later levels. The principal proposed criterion is

$$\mathcal{L}_{\mathrm{tc}} = \sum_{k=1}^{K} \alpha_k \, \mathbb{E}_{u, C_{<k}} \mathrm{TC}(C_{k,1:n_k} \mid C_{<k}, u), \tag{6}$$

where TC is defined in Section 5. For practical codebooks, the target rate is not merely $\sum_k n_k \log_2 V_k$: serialization, fallback, geometry and entropy coding must be measured separately. A fixed codec can also be tested with an adaptively chosen level ordering; the proposed *learned* hierarchy introduces a distinct optimization problem.

4.5 Generative prior synthesis

After the codec is held fixed, SCALE trains an explicitly prompt-conditioned shared level-wise prior of the form (2). Prompt/continuation separation, causal prompt encoding, output-length strategy and fallback encoding are part of the interface. The existing reference prior lacks a general user-prompt cross-attention interface; this is an explicit development requirement, not an implemented property [2]. Teacher-forced code likelihood is supplemented by free-running sequence evaluation and conditioning corruption, since errors in earlier levels propagate to later levels [13].

4.6 Adaptive verification and failure handling

At each stage the compiler runs bounded tests on independent data. If the representation residual, calibration sensitivity, level conditional dependence, or output KL is too large, it may (a) increase canonical rank or code budget; (b) reallocate levels or dependence-sensitive codes; (c) choose restricted distillation for specific modules; or (d) report that the requested resource envelope is infeasible. An honest failure report is a valid compiler outcome. This distinguishes *adaptivity* from a blanket claim of lossless compression.

5 Mathematical Feasibility and Limitations

The following propositions give *conditional* correctness statements for parts of SCALE. They do not prove that actual independently trained Transformers obey their premises.

Proposition 1 (Exact transfer through an invariant canonical subspace)**.** *Let $A_s \in \mathbb{R}^{d_s \times r}$ have full column rank, and let $P_s \in \mathbb{R}^{r \times d_s}$ satisfy $P_s A_s = I_r$. Suppose teacher operator $F_s : \mathbb{R}^{d_s} \to \mathbb{R}^{d_s}$ and canonical operator $M : \mathbb{R}^r \to \mathbb{R}^r$ obey*

$$F_s(A_s z) = A_s M(z) \qquad \text{for every } z \in \mathcal{Z}.$$

Then the compiled canonical operator $\widehat{M}_s(z) = P_s F_s(A_s z)$ equals $M(z)$ for all $z \in \mathcal{Z}$, independently of d_s. For linear $F_s(h) = W_s h$ and $M(z) = W_* z$, the equality is $W_* = P_s W_s A_s$.

Proof. Substitution yields $P_s F_s(A_s z) = P_s A_s M(z) = I_r M(z) = M(z)$. In the linear case substitute $F_s(h) = W_s h$ and identify the matrices. □

Interpretation. The theorem establishes an exact, width-independent *rule* only on an invariant subspace, with the maps A_s, P_s supplied or correctly recovered. It says nothing about obtaining these maps cheaply, discovering an invariant subspace in real LLMs, or transferring a nonlinear attention block outside that subspace. This is a sufficiency result, not a general algorithmic guarantee.

Proposition 2 (One-step predictive bound)**.** *Let $h_s = A_s z + \eta_s$, $P_s A_s = I_r$, and let a code/decoder reconstruction satisfy $\|Q(P_s h_s) - P_s h_s\| \le \epsilon_Q$ in a common norm. Suppose there is an L-Lipschitz logit function $g : \mathbb{R}^r \to \mathbb{R}^V$ (from the latent norm to ℓ_∞), with teacher logits ℓ_T satisfying $\|\ell_T - g(z)\|_\infty \le \epsilon_T$, and student logits ℓ_S satisfying $\|\ell_S - g(Q(P_s h_s))\|_\infty \le \epsilon_D$. Define*

$$\Delta = \epsilon_T + L(\|P_s\|\,\|\eta_s\| + \epsilon_Q) + \epsilon_D. \tag{7}$$

Then the softmax next-token distributions obey

$$D_{\mathrm{KL}}\big(\mathrm{softmax}(\ell_T) \,\|\, \mathrm{softmax}(\ell_S)\big) \le 2\Delta.$$

Proof. Since $P_s h_s = z + P_s \eta_s$, the triangle inequality gives $\|Q(P_s h_s) - z\| \le \epsilon_Q + \|P_s\|\,\|\eta_s\|$. Lipschitzness and the assumed logit errors yield $\|\ell_T - \ell_S\|_\infty \le \Delta$. If every coordinate of logits a, b differs by at most Δ, the log-partition functions $\log \sum_i e^{a_i}$ and $\log \sum_i e^{b_i}$ differ by at most Δ. Therefore $\log(\mathrm{softmax}(a)_i / \mathrm{softmax}(b)_i) \le 2\Delta$ for each i. Taking expectation under $\mathrm{softmax}(a)$ proves the claim. □

Caution. Proposition 2 concerns matched-vocabulary next-token logits with a compatible output representation. It neither guarantees small $\epsilon_T, \epsilon_Q, \epsilon_D$ nor bounds unconstrained long-sequence free-running quality. A uniform-in-size bound follows only *if* those errors, L, and $\|P_s\|$ remain uniformly bounded across s.

Proposition 3 (Within-level factorization gap)**.** *Fix a preceding-level context $b = (C_{<k}, u, G)$ and a conditional distribution $P(C_{k,1:n_k} \mid b)$. Among all conditionally independent per-position priors $Q(C_k \mid b) = \prod_j q_j(C_{k,j} \mid b)$,*

$$\begin{aligned}\min_Q D_{\mathrm{KL}}(P(\cdot \mid b) \| Q(\cdot \mid b)) &= \sum_{j=1}^{n_k} H_P(C_{k,j} \mid b) - H_P(C_{k,1:n_k} \mid b) \\ &= \mathrm{TC}_P(C_{k,1:n_k} \mid b),\end{aligned}$$

with optimum $q_j = P(C_{k,j} \mid b)$ for each j. Averaging over contexts yields the expected conditional total correlation.

Proof. Expand the divergence as $-H_P(C_k \mid b) - \sum_j \mathbb{E}_P \log q_j(C_{k,j} \mid b)$. Each cross-entropy term is minimized by the corresponding marginal $P(C_{k,j} \mid b)$ and then equals its entropy. The sum minus the joint entropy is the definition of total correlation. □

This identity is a specialization of a known information-theoretic result about parallel sampling [18], not a novel theorem claimed by SCALE. It gives a principled signal for designing q_ϕ: move highly shared dependencies into earlier levels so that later parallel decisions are closer to conditionally independent.

Concrete counterexample. If $P(C_1C_2 = 00) = P(11) = 1/2$, all other outcomes being impossible, a factorized one-round model using correct marginals assigns probability $1/4$ to each pair. Its irreducible divergence is $\log 2$ nats. Teacher feature fidelity by itself cannot remove this gap; the hierarchy must convey the shared bit earlier or change the sampling factorization.

Proposition 4 (Finite-code capacity obstruction)**.** *Let a deterministic codec produce at most M distinct codewords and let a decoder depend on the encoded input only through that codeword (with all other conditioning fixed). If a finite set of target continuations contains $N > M$ pairwise distinct items that must all be reconstructed exactly, exact reconstruction of all N continuations is impossible.*

Proof. By the pigeonhole principle, two distinct continuations receive the same codeword, and the deterministic decoder returns the same reconstruction for both. It therefore cannot reproduce both exactly. □

For codebook sizes V_k and level lengths n_k, a naive upper bound is $M \le \prod_k V_k^{n_k}$, or $\log_2 M \le \sum_k n_k \log_2 V_k$. This ignores geometry/fallback side channels and may overstate realizable distinct reconstructions. The proposition refutes universal lossless conversion under fixed finite capacity; it does *not* claim that high-quality approximate generation necessarily requires exact data reconstruction.

Lemma 1 (Calibration-only indistinguishability)**.** *For any finite calibration set A within a larger domain $\mathcal{X}$, there exist two teacher functions agreeing on every point of A but disagreeing at some point of $\mathcal{X} \setminus A$. Any compiler decision using only teacher input–output responses on A is identical for both teachers and cannot guarantee exact behavior on all of $\mathcal{X}$ for both.*

Proof. Choose any $x^* \notin A$. Define two functions to agree on A and disagree only at x^*. They are observationally indistinguishable to an algorithm whose teacher observations are restricted to A. The algorithm outputs the same hypothesis for both, which cannot match both at x^*. □

The lemma applies to *calibration-only* evidence. A compiler that also inspects complete weights has additional information; nevertheless, the lemma explains why held-out evaluation and explicit model-class assumptions are necessary.

6 Training Objectives and Practical Dependence Estimation

A theoretically explicit objective is not automatically computable. For a deterministic codec $q_\phi(x, u)$, teacher sampling induces the *pushforward* code distribution

$$P_{s,\phi}(c \mid u) = \sum_{x:q_\phi(x,u)=c} p_s(x \mid u). \tag{8}$$

The codec and therefore this target distribution change when ϕ changes; joint optimization can create unstable moving labels. SCALE consequently proposes controlled alternation: (1) collect teacher-generated samples and teacher activations; (2) update or initialize the codec with a fixed teacher and a selected hierarchy; (3) freeze the codec and fit the level prior; (4) run free-generation and update the design only after diagnostic evaluation. This resembles the staged tokenizer/prior separation in original NCM [13].

Representation fidelity. With a teacher–student alignment map P_s and masked valid positions $\mathcal{M}$, define

$$\mathcal{L}_{\text{rep}} = \mathbb{E}_{(u,x)} \sum_{t \in \mathcal{M}} \|\hat{z}_t(x, u) - P_s h_{s,t}(x, u)\|_2^2 / |\mathcal{M}|.$$

Supplement it with teacher/student logits, decoding accuracy and task probes; an activation MSE or a high CKA score alone cannot establish knowledge preservation [12].

Rate–dependence objective. For each level, $\mathcal{L}_{\text{rate}}$ penalizes measured entropy-coded bits, including masks, length/fallback indicators and code metadata. The proposed level objective is

$$\begin{aligned}\mathcal{L}_{\text{codec}} = \lambda_{\text{rec}}\mathcal{L}_{\text{reconstruction}} + \lambda_{\text{rep}}\mathcal{L}_{\text{rep}} \\ + \lambda_{\text{tc}} \sum_k \alpha_k \mathbb{E}_b \text{TC}(C_k \mid b) + \lambda_{\text{rate}}\mathcal{L}_{\text{rate}}.\end{aligned} \tag{9}$$

Exact high-dimensional TC may be intractable. Under adequately expressive and well-fitted joint and factorized conditional density models, the difference in their optimal negative log-likelihoods estimates conditional total correlation. With finite critics it is only a *surrogate*; it is not generally an unbiased certified bound. The evaluation must compare estimated dependence against independent estimators on small enumerable toy problems and report critic capacity sensitivity. A stop-gradient or alternating critic update prevents trivially gaming a weak critic from being mistaken for information reduction.

Prior and behavior. For frozen codes $c = q_\phi(x, u)$, the prior loss is

$$\mathcal{L}_{\text{prior}} = -\mathbb{E}_{(u,x)} \sum_{k,j} \log q_{\omega,k,j}(c_{k,j} \mid c_{<k}, u, G), \tag{10}$$

normalized over valid positions. Teacher logit matching or sampled-response distillation can be added if a common output tokenization is supported, but discrete-code CE is not the same objective as teacher next-token KL. Full generated text must be measured with actual sampling and decoding, not just teacher-forced code accuracy.

Selective transfer policy. For every candidate subgraph, allocate one of four statuses: EXACT (symbolically justified equivalence on a specified domain), ALIGNED (calibrated map, bounded empirical residual), DISTILL (trainable correction with a recorded budget), or REJECT (architecture/resource incompatibility). Exact preservation must be a proof about a *function*, not merely a low finite-sample numerical discrepancy. Retraining should be separately counted, including optimizer state and teacher inference.

7 Proposed Algorithm and Adaptive Control

Algorithm 1: SCALE (proposed, not yet implemented)

1. **Inspect** teacher architecture, weights, tokenizer, license, provenance, and target resource specification.
2. **Collect** training-only calibration activations, logits and sampled continuations, with independent held-out calibration and evaluation splits.
3. **Align** layerwise source activations to a frozen canonical core; fit model-specific maps; quantify alignment uncertainty and residuals.
4. **Classify** candidate operators as exact, aligned, distill or reject; record mathematical certificates or empirical residuals.
5. **Design** RVQ levels and codebooks by optimizing reconstruction, predictive fidelity, measured rate and a conditional-TC surrogate.
6. **Freeze codec and fit prior** with explicit prompt conditioning, mask/length contracts and controlled conditioning corruption.
7. **Evaluate** free-running outputs, conditional dependence, transfer fidelity, total FLOPs, latency and memory on a sequestered split.
8. **Adapt or reject**: if constraints fail, increase latent/code capacity, change scheduling or allocate selective distillation, subject to budget $\mathcal{R}$; otherwise emit compiled checkpoint plus an audit report.

A practical compiler needs termination conditions (maximum retries and resources), immutable dataset and model hashes, and reports that distinguish hard constraints from soft improvement. In particular, "no full retraining" does not mean constant total compilation cost: teacher inspection and inference grow with teacher size, and calibrating a map may scale with hidden width.

8 Existing Repository: Evidence and Proposed Extension

The public codebase `https://github.com/fbalensiefer/Next-Concept-Modeling-for-TPU` is titled *Next-Concept Modeling for TPU*, authored by Fabian Balensiefer and released as version 0.1.0 on November 11, 2026 under the MIT license [3]. It credits NCM to Kumbong et al. [13]. The accompanying manuscript [2] and repository source define the following *implemented* functionality:

- `src/ncm/tokenizer.py`, `rvq.py` and `grids.py`: masked convolutional encoder/decoder, multiscale residual VQ, explicit EMA codebook state and fixed-grid alignment;
- `src/ncm/generator.py`: a shared Transformer scanned over levels with earlier-level-only conditioning and parallel within-level predictions;
- `src/ncm/training.py`, `losses.py`, `sharding.py`: functional optimizer/EMA transitions, reconstruction/prior objectives, runtime phase control, and explicit `dp`/`fsdp` placement;
- `tests/`, `examples/train.py`: software-correctness tests and three synthetic training steps.

The validation record for the November 11 snapshot reports 11 passing CPU tests, including a virtual 2×2 data/model-mesh check against an unsharded step. It does *not* demonstrate a trained language model, semantic quality, a TPU benchmark, a scalable Rust pipeline, or cross-architecture transfer. The current prior has no general prompt cross-attention interface,

and codebook/shape contracts are fixed per compiled geometry [2]. **None of the modules proposed below should be described as already implemented.**

Table 1: Proposed SCALE modules and integration points; all are future work.

Proposed module	Responsibility / interface
`scale/teachers.py`	Frozen teacher logits, hidden states, tokens, revision IDs and licensing checks.
`scale/align.py`	Multi-layer canonical alignment, ridge/transport maps, diagnostics and holdout tests.
`scale/operators.py`	Typed operator compatibility, exact certificates, residual analysis and selective adaptation.
`scale/dependence.py`	Conditional TC estimators/critics, bias checks and uncertainty reporting.
`scale/hierarchy.py`	Rate–distortion–dependence optimization and static-shape schedule compiler.
`scale/prior.py`	Prompt-conditioned NCM level prior and corruption-robust sampling.
`scale/compiler.py`	Audited compile policy, adaptive budget allocation, checkpoint manifests.
`scale/evaluate.py`	Held-out fidelity, free generation, GPU/TPU compute accounting and baselines.

Execution implications. The underlying JAX implementation pads all levels to a common length; dense resampling and repeated level-wise attention can dominate cost. For K levels, length L, hidden width H, J attention blocks, batch B, and codebook size V, its source analysis includes roughly $O(KJB(L^2H + LH^2) + KBLHV)$ prior work as well as dense resampling costs [2]. Thus fewer sequential levels do not automatically imply lower wall-clock latency. SCALE must compare actual synchronized throughput, peak memory and compile time against autoregressive teachers and NCM baselines.

9 Evaluation Protocol and Preregistered Falsification

9.1 Research questions and hypotheses

RQ1 (size transfer): Can a canonical compiler core calibrated on small Pythia teachers generalize to held-out larger teachers without being retrained? **H1:** It reduces total additional compute versus a parameter-matched NCM from-scratch baseline without unacceptable held-out quality loss.

RQ2 (dependency awareness): Does minimizing estimated conditional TC at a fixed code budget improve free-running quality/latency compared with reconstruction-only RVQ? **H2:** It improves the quality–parallel-steps Pareto frontier on unseen data, beyond a decoder/prior-capacity explanation.

RQ3 (selective reuse): Does exact/aligned transplantation followed by selective distillation outperform direct distillation with the same training resource budget? **H3:** It reduces adaptation FLOPs without degrading output fidelity at equal target size.

RQ4 (integration): Does the full combination have positive interaction beyond its individual components? **H4:** A factorial three-factor ablation identifies an interaction in quality–cost performance, not only additive improvements.

9.2 Model and data ladder

Begin with the controlled Pythia family [4]: develop the core with 70M/160M sizes and fixed anchor prompts; lock the core before evaluation at 410M and approximately 1B; only then test a different model family with compatible access and licensing. Do not use scaling checkpoints of a nominally "held-out" teacher to tune the core. Fix train/calibration/validation/test splits by hash; explicitly control vocabulary bridges, prompt lengths, continuation lengths and sampling temperatures. Teacher-generated outputs are expensive and should be cached with their exact sampling metadata.

9.3 Factorial baselines

At minimum compare (i) NCM from scratch under matched data/parameter budgets, (ii) ordinary sequence/logit distillation, (iii) teacher-initialized codec with no TC regularizer, (iv) TC-aware codec without canonical cross-size transfer, (v) selective reuse without TC, and (vi) full SCALE. A 2^3 factorial design for alignment transfer, TC-aware hierarchy, and operator reuse distinguishes main effects from interactions. Strong published cross-architecture alignment/merging precedents should be reimplemented or benchmarked where technically comparable [5, 8, 14]. Since the latter primarily address merging, do not claim a direct generative-head equivalence without an appropriate adaptation.

9.4 Metrics and resource accounting

Report teacher/student token-level cross-entropy and KL on matched tokenizations, original output-space generative evaluations, reconstruction exact-match, code utilization and perplexity, estimated and small-domain exact TC, hard serialized bits, free-running error propagation, generated-token quality metrics with stated limitations, and robustness across seeds. Compute accounting includes teacher forward FLOPs, feature extraction, alignment search, critic training, distillation, failed compilation attempts, JAX compilation, checkpoint I/O, and evaluation. Report *total incremental cost* as well as inference latency/throughput, energy or accelerator-hours where available, peak device RAM, and quality–cost Pareto frontiers. Model quality and speed must be assessed on real accelerators, not inferred from unit tests or asymptotic complexity.

Table 2: Predefined failure observations; these falsify or narrow stated hypotheses.

Hypothesis	**Failure or narrowing condition**
H1: cross-size generality	Fixed core loses held-out fidelity at larger scales; teacher-specific retraining dominates savings.
H2: dependency benefit	Estimated TC decreases but independently estimated dependence or free-running generation does not improve.
H3: selective reuse	No significant gain at equal *total* additional FLOPs, or transplanted parts destabilize training.
H4: synergy	Full SCALE has no positive interaction or is Pareto-dominated by simpler baselines.
Feasibility envelope	Quality targets require code rate, memory, sequential depth or compute above the declared budget.

All thresholds must be fixed *before* examining the test data. The phrase "less retraining" requires a quantitative reference; a proposal target such as a $\geq 30\%$ reduction in total incremental training FLOPs at matched quality may be preregistered, but is not a measured result. Statistical intervals should cover multiple seeds, prompts, and held-out domains, with unsuccessful trials reported.

10 Threats to Validity and Research Integrity

Non-identifiability and negative transfer. Functional spaces are not unique; affine alignment can preserve certain probes while erasing rare or compositional capabilities. Bigger teachers may acquire features outside a fixed canonical rank. The adaptive response should be to increase capacity or reject the transfer rather than claim universal scale invariance.

Optimization and dependence estimators. Learning a hierarchy to minimize measured TC can exploit weak density critics or reallocate entropy into harder-to-estimate coordinates; conditional entropy estimators should be calibrated on synthetic distributions where ground truth is enumerable. Low teacher-forced code loss need not predict free-running behavior.

Prompt conditioning and tokenization. The base repository's unconditioned first-level path does not provide a chatbot interface. Prefix causality, tokenizer mismatch, byte fallback serialization, output length and stop behavior require separate implementation and validation. Source-side teacher tokenizer compatibility cannot be assumed.

Theoretical limits. Proposition 1 is conditional on a shared invariant structure; Proposition 2 is a local logit bound under common output semantics; Proposition 3 exposes a factorization cost but does not show that trainable RVQ can eliminate it; Proposition 4 excludes a universal finite-code lossless transfer; and Lemma 1 invalidates universal assertions from finite black-box calibration alone. No statement here proves that real Transformers possess a sufficiently stable, low-rank canonical space.

Novelty and reproducibility. Prior work already covers cross-size stitching, latent weight merging, transport alignment, RVQ, and the conditional-TC identity [5, 8, 11, 14, 18]. The prospective novelty is the specific audited, dependency-aware *generative compilation* algorithm and demonstrated interaction between components, not each ingredient separately. A preprint should accompany source revision hashes, frozen experimental protocol, dataset manifests, exact inference configurations, negative results and a public reproducibility checklist. This paper itself reports neither new experiments nor a comprehensive patent novelty search.

11 Discussion and Conclusion

A neural model is more than a collection of tensors: its knowledge is distributed over computations, its representation is not uniquely identified by coordinates, and its generative factorization imposes a specific dependency structure. SCALE therefore treats cross-architecture transfer as the compilation of a *functionally tested intermediate representation into an information-aware NCM hierarchy*, followed by the synthesis of a compatible generative prior and selective adaptation. The mathematical results specify sufficient conditions for exact operator transfer and conditional output fidelity, while finite code capacity and within-level conditional dependence rule out universal lossless claims.

The next decisive experiment is not a large end-to-end benchmark but a discriminating smaller test: align two small Pythia teachers to a fixed canonical interface, quantify how well a held-out larger teacher maps to it, and test whether a dependency-aware RVQ hierarchy improves the frozen-code prior over a reconstruction-only one. If this fails under adequate controls, the central SCALE hypothesis should be revised or rejected rather than protected by additional unbounded

training. The existing JAX/Flax repository is a transparent starting point for that test; it is not evidence that the proposed compiler already works.

Code and Source Availability

Underlying systems project: public GitHub repository (Fabian Balensiefer, release 0.1.0, November 11, 2026; MIT license). The accompanying source paper [2] describes the existing implementation and its CPU-only checks. SCALE-specific code, pretrained compiler weights, NCM transfer checkpoints, and empirical comparisons are *proposed future artifacts*; they are not presently claimed as available.

Author Contributions and Declaration

Fabian Balensiefer is the named author of this research concept and the existing systems implementation. The technical development of the proposed compiler, mathematical assumptions and experimental protocols in this manuscript should be treated as research hypotheses pending implementation and peer review. The underlying Next-Concept Modeling method is explicitly credited to Kumbong and colleagues. No third-party model performance result is represented as a SCALE result.

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